

Incident Report: nginx Fails to Start After Config Syntax Error

Scenario 1 • Linux Web Server Troubleshooting Labs

Summary

nginx failed to start after a restart following a configuration change. A missing semicolon in `/etc/nginx/sites-enabled/default` caused the config test to fail, so systemd refused to start the service. The service was restored by correcting the syntax error and verified with both a service status check and an HTTP request.

This was a controlled lab exercise. The fault was introduced deliberately to practice diagnosis and recovery.

Environment

- Debian 13 virtual machine (VirtualBox), hostname `soc-lab`
- nginx web server, managed by systemd
- Tools used: `systemctl`, `journalctl`, `nginx -t`, `curl`, `sed`, `grep`

Timeline (EDT)

Time	Event
19:37	Baseline check: nginx active (running), <code>curl -I</code> returns 200 OK
20:02:13	Removed the semicolon on line 22 and restarted nginx. Restart failed
20:02:13	Journal records <code>invalid parameter "listen"</code> and Failed to start nginx.service
20:36:27	<code>nginx -t</code> reports the error at <code>default:23</code>
After 20:36	Semicolon restored on line 22. <code>nginx -t</code> passes
20:52:54	nginx restarted successfully. <code>curl -I</code> returns 200 OK

Symptom

`sudo systemctl restart nginx` failed with: *Job for nginx.service failed because the control process exited with error code.* The service state was failed (Result: exit-code) and the website was unavailable.

Investigation

- 1. Check service state.** `systemctl status nginx --no-pager` showed Active: failed and that the ExecStartPre step (`nginx -t`) exited with status=1/FAILURE. systemd runs a config test before starting nginx; the test failed, so nginx never started.
- 2. Read the logs.** `sudo journalctl -u nginx -n 20 --no-pager` showed [emerg] invalid parameter "listen" and the message that the configuration file test failed.
- 3. Run the config test directly.** `sudo nginx -t` named the file and line:
`/etc/nginx/sites-enabled/default:23.`

4. Inspect the config. `sed -n '20,24p' /etc/nginx/sites-enabled/default` showed line 22 (`listen 80 default_server`) was missing its closing semicolon.

Root Cause

A missing semicolon at the end of line 22 (`listen 80 default_server`). nginx read lines 22 and 23 as a single directive, so it only noticed a problem when it reached the second `listen` on line 23. **The error was reported one line after the actual mistake**, so when nginx points to a line, the line above it is worth checking too.

Resolution

Restored the semicolon on line 22:

```
sudo sed -i '22s$/;/' /etc/nginx/sites-enabled/default
```

Confirmed with `sed -n '20,24p'` that the line ends with a single semicolon, then validated and restarted.

Verification

- `sudo nginx -t`: syntax is ok and test is successful
- `systemctl status nginx --no-pager`: active (running), with the `ExecStartPre` config test exiting `status=0/SUCCESS`
- `curl -I http://127.0.0.1: HTTP/1.1 200 OK`

Prevention

- Run `sudo nginx -t` before every restart or reload.
- When nginx reports an error on a line, also check the line above it.
- Back up config files before editing, and store backups **outside** `sites-enabled/` (nginx loads every file in that directory, so a backup left there can cause a duplicate server error).
- In production, use change control and a config validation step so a bad edit never reaches a restart.

Evidence

1. Baseline: nginx healthy before the change

```
ETag: "6abaf932-267"
Accept-Ranges: bytes

mohammed@soc-lab:~$ systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-10-04 19:16:17 EDT; 20min ago
     Invocation: cf83cd363da44952938d12329ce06cf4
       Docs: man:nginx(8)
    Process: 1788 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 1789 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1791 (nginx)
      Tasks: 2 (limit: 2301)
     Memory: 2.3M (peak: 2.5M)
        CPU: 40ms
      CGroup: /system.slice/nginx.service
              └─1791 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
                 └─1792 "nginx: worker process"

mohammed@soc-lab:~$ curl -I http://127.0.0.1
HTTP/1.1 200 OK
Server: nginx
Date: Sun, 04 Oct 2026 23:37:48 GMT
Content-Type: text/html
Content-Length: 615
Last-Modified: Mon, 28 Sep 2026 23:33:06 GMT
Connection: keep-alive
ETag: "6abaf932-267"
Accept-Ranges: bytes

mohammed@soc-lab:~$
```

2. Config broken: line 22 missing its semicolon

```
mohammed@soc-lab:~$ sed -n '20,24p' /etc/nginx/sites-enabled/default
#
server {
    listen 80 default_server
    listen [::]:80 default_server;

mohammed@soc-lab:~$ _
```

3. Restart fails

```
mohammed@soc-lab:~$ sudo systemctl restart nginx
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
mohammed@soc-lab:~$
```

4. Service status: failed, config test exited with an error

```
mohammed@soc-lab:~$ systemctl status nginx --no-pager
* nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Sun 2026-10-04 20:02:13 EDT; 30min ago
     Duration: 45min 55.560s
  Invocation: 805715ee33f24305b1770934a26bb5dd
       Docs: man:nginx(8)
    Process: 1950 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=1/FAILURE)
   Mem peak: 1.7M
        CPU: 13ms
mohammed@soc-lab:~$ _
```

5. Journal: invalid parameter "listen"

```
mohammed@soc-lab:~$ sudo journalctl -u nginx -n 20 --no-pager
[sudo] password for mohammed:
Oct 04 18:57:45 soc-lab systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
Oct 04 18:57:45 soc-lab systemd[1]: nginx.service: Deactivated successfully.
Oct 04 18:57:45 soc-lab systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
Oct 04 18:57:45 soc-lab systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 04 18:57:45 soc-lab nginx[1676]: 2026/10/04 18:57:45 [emerg] 1676#1676: invalid parameter "listen" in /etc/nginx/sites-enabled
Oct 04 18:57:45 soc-lab nginx[1676]: nginx: configuration file /etc/nginx/nginx.conf test failed
Oct 04 18:57:45 soc-lab systemd[1]: nginx.service: Control process exited, code=exited, status=1/FAILURE
Oct 04 18:57:45 soc-lab systemd[1]: nginx.service: Failed with result 'exit-code'.
Oct 04 18:57:45 soc-lab systemd[1]: Failed to start nginx.service - A high performance web server and a reverse proxy server.
Oct 04 19:16:17 soc-lab systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 04 19:16:17 soc-lab systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
Oct 04 20:02:13 soc-lab systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
Oct 04 20:02:13 soc-lab systemd[1]: nginx.service: Deactivated successfully.
Oct 04 20:02:13 soc-lab systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
Oct 04 20:02:13 soc-lab systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 04 20:02:13 soc-lab nginx[1950]: 2026/10/04 20:02:13 [emerg] 1950#1950: invalid parameter "listen" in /etc/nginx/sites-enabled
Oct 04 20:02:13 soc-lab nginx[1950]: nginx: configuration file /etc/nginx/nginx.conf test failed
Oct 04 20:02:13 soc-lab systemd[1]: nginx.service: Control process exited, code=exited, status=1/FAILURE
Oct 04 20:02:13 soc-lab systemd[1]: nginx.service: Failed with result 'exit-code'.
Oct 04 20:02:13 soc-lab systemd[1]: Failed to start nginx.service - A high performance web server and a reverse proxy server.
mohammed@soc-lab:~$ _
```

6. nginx -t points to default:23

```
mohammed@soc-lab:~$ sudo nginx -t
2026/10/04 20:36:27 [emerg] 2097#2097: invalid parameter "listen" in /etc/nginx/sites-enabled/default:23
nginx: configuration file /etc/nginx/nginx.conf test failed
mohammed@soc-lab:~$
```

7. After the fix: config test passes

```
mohammed@soc-lab:~$ sudo sed -i '22s/$/;/' /etc/nginx/sites-enabled/default
mohammed@soc-lab:~$ sed -n '20,24p' /etc/nginx/sites-enabled/default
#
server {
    listen 80 default_server;
    listen [::]:80 default_server;

mohammed@soc-lab:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
mohammed@soc-lab:~$ _
```

8. Recovery: service running and returning 200 OK

```
mohammed@soc-lab:~$ sudo systemctl restart nginx
mohammed@soc-lab:~$ systemctl status nginx --no-pager
• nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-10-04 20:52:54 EDT; 13s ago
     Invocation: 0ded2b71cc904f3494b4328cdacaf52f
       Docs: man:nginx(8)
    Process: 2181 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 2184 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2185 (nginx)
      Tasks: 2 (limit: 2301)
     Memory: 2.3M (peak: 2.5M)
        CPU: 39ms
    CGroup: /system.slice/nginx.service
            └─2185 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─2187 "nginx: worker process"

mohammed@soc-lab:~$ curl -I http://127.0.0.1
HTTP/1.1 200 OK
Server: nginx
Date: Mon, 05 Oct 2026 00:53:21 GMT
Content-Type: text/html
Content-Length: 615
Last-Modified: Mon, 28 Sep 2026 23:33:06 GMT
Connection: keep-alive
ETag: "6abaf932-267"
Accept-Ranges: bytes

mohammed@soc-lab:~$
```